

EFFECTIVENESS OF ARTIFICIAL INTELLIGENCE IN TECHNICAL WRITING
SKILLS: ACADEMIC PERFORMANCE OF SECOND YEAR BACHELOR OF
SCIENCE IN BIOLOGY OF AKLAN STATE UNIVERSITY

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CHAPTER I

INTRODUCTION

Background of the Study

The adoption of Artificial Intelligence (AI) technologies is profoundly impacting the dynamics of scientific communication. Technical writing, one of the crucial communication disciplines in science that requires precision and exactness, is no exception to these developments (Beilfuss et al. 685). Technical writing within the biological sciences is crucial because it helps articulate complex scientific concepts, techniques, and findings for diverse audiences through reports, research papers, and articles (Alian and Awajan 1).

The advent of AI powered writing technology, such as NLG models, grammar checkers, paraphrasers, and research assistants, has transformed the possibilities for automating the writing process. As Raja and Nagasubramani noted, these tools promise increased productivity, improved language standards, easier content creation, smarter handling of research and citations, among others (685-688). At the same time, Aklan State University... concerns AI technologies stimulate, such as the balanced development of fundamental writing skills, are quite difficult to alleviate. Some concerns include dependency on AI tools, plagiarism, authorship issues, and academic dishonesty (Zawacki-Richter et al. 103611).

It is important for students pursuing their Bachelor of Science in Biology to understand the impact of AI on their technical writing skills when it comes to developing scientific literacy and communication skills. Considering how these students start using these tools, it is important to examine what effect, if any, this

interaction has on their ability to independently create, express and convey scientific information (Tran 1-12). This study examines whether AI tools assist in understanding technical writing concepts, or whether they hinder the development of critical thinking, research and independent writing skills needed for their future career as a biologist.

This research seeks to explore the implications of integrating artificial intelligence within the Aklan State University Biology program, aiming to offer significant insights into its effective incorporation into the curriculum. The objective is to enhance, rather than replace, the cultivation of strong technical writing abilities in aspiring biologists. The results will ultimately contribute to teaching methodologies, influence curriculum design, and assist in the creation of thorough guidelines for the responsible use of AI in academic writing for biology students.

Statement of the Problem

This research seeks to assess the impact of artificial intelligence (AI) on the technical writing abilities of second-year Bachelor of Science in Biology students at Aklan State University. The emphasis is on examining how AI tools impact various aspects of technical writing, such as clarity, conciseness, accuracy, and overall effectiveness, for this specific group of students. The study will explore whether utilizing AI enhances or hinders the development of crucial technical writing abilities among these students. The absence of studies examining the effect of AI on technical writing abilities in the realm of biology education drives this research.

Specifically, the following questions are addressed on this study:

1. What is the extent of using AI tools in enhancing or diminishing the effectiveness of coherence and conciseness in technical writing achieved by second year Bachelors of Science in Biology Students in Aklan State University?
2. In what ways does the utilization of AI impact the efficiency in performing scientific literacy with regards to the written work addressing the appropriate audience?
3. What are the advantages and the disadvantages AI tools for technical writing might bring to the undergraduates of Bachelor of Science in Biology at Aklan State University in terms of skill acquisition, learning achievement, and overall learning performance?

Theoretical Framework

According to Walberg's (1991) theory of educational productivity, it addresses the effects on learning that affect the student's academic performance. It is an investigation of academic accomplishment in which Walberg employed a range of approaches to determine the elements that influence a student's academic performance.

The adoption of artificial intelligence applications in higher education plays an important role in the improvement of the quality of education and learning practices and overcomes many educational issues. The purpose of this study is to examine the effects of artificial intelligence in technical writing skills influencing academic performance of 2nd year Bachelor of Science in Biology of Aklan State University by using it in the educational process. The use of artificial intelligence has a positive and significant effect on learning performance.

In academic writing instruction at the university level, AI entails using computer systems and technologies to perform tasks typically associated with human Intelligence and cognitive abilities (Leoste et al., 2021; Kamalov et al., 2023). Specifically for academic writing courses, AI can encompass tools, software, and language models designed to assist students and instructors with various aspects of the writing process (Akgun & Greenhow, 2022; Fyfe, 2023; Gardner et al., 2021). Considering the challenges students experience, AI has emerged as central to enhancing academic writing courses.

Additional theories explored in AI integration in academic courses include cognitive load theory, activity theory, critical pedagogy, and ecological systems theory. Cognitive load theory, developed by Sweller (1988), examines the mental effort required for learning, helping understand whether AI enhances or hinders students' learning experience (Berssanette & de Francisco, 2022). Activity Theory, developed by Vygotsky and expanded by Leont'ev, focuses on individual interactions and their social and cultural context (Li, 2023). Researchers have applied activity theory to understand the impact of AI tools on academic writing activities, shaping the roles of students, instructors, and tools within the writing process (Berssanette & de Francisco, 2022).

Critical Pedagogy is rooted in the works of Paulo Freire and focuses on questioning and challenging established norms (Pronzato & Markham, 2023). Critical pedagogy is essential in examining the socio-political implications of relying on AI tools, including access, power dynamics, and bias. Ecological Systems Theory, developed by Urie Bronfenbrenner, has been applied to examine how the introduction of AI tools influences the writing ecosystem, including interactions with peers, instructors, and the broader educational context (Sajjad et al., 2023). These theories provide lenses through which researchers and educators can analyze and understand the multifaceted impact of integrating AI tools into academic writing courses.

Conceptual Framework

This research is based on the interrelationship between the technological advancement and educational development through which the influence of the artificial intelligence (AI) tools on the technical writing skills of the second year Bachelor of Science in Biology students of Aklan State University is studied. It takes a constructivist perspective, which positions that students learn best through involvement and knowledge construction, and examines if AI tools either enhance or hinder this learning process within the realm of scientific writing.

The conceptual basis of this study rests on three core constructs: "AI tool usage", "technical writing performance", and "cognitive writing development". AI tool usage includes grammar checkers, paraphrasing tools, content generators, and citation managers. Technical writing performance refers to measurable aspects such as coherence, clarity, accuracy, and structure. Cognitive writing development addresses the student's independent abilities in critical thinking, idea formulation, and audienceaware communication.

This conceptual framework suggests that the inclusion of AI tools can be both scaffold and barrier. Although AI is capable of providing instant feedback and support that improves the quality of the output, excessive use could dilute the acquisition of necessary writing skills. The conceptual framework thus warrants an assessment of AI's dual functions both as an enhancer and as a likely clutch in academic writing.

This model assists in directing the research in identifying how students utilize AI tools, what skills are acquired or lost in the process, and how these engagements

affect learning outcomes for technical writing. The research therefore seeks to inform equitable strategies for integrating AI into education to ensure maximum benefits against its determinants.

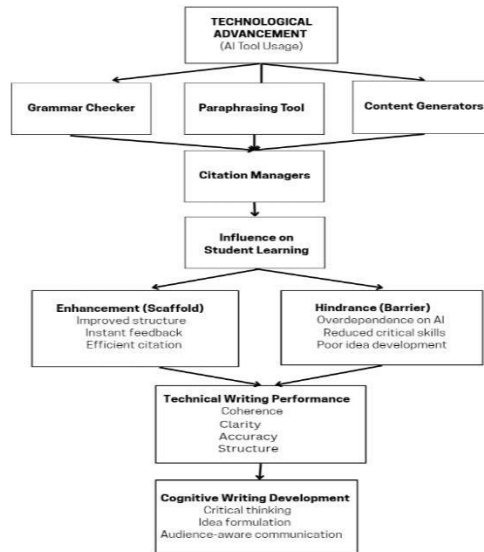


Figure 1. Conceptual framework illustrating the relationship between artificial intelligence (AI) tool usage and its dual influence as both an enhancer (scaffold) and hindrance (barrier) on the technical writing performance and cognitive writing development of second-year Bachelor of Science in Biology students at Aklan State University.

Significance of the Study

The study aims to determine The Effects of Artificial Intelligence in Technical Writing Skills on of 2nd Year Bachelor of Science in Biology of Aklan State University as it contributes valuable knowledge. On an era where AI tools like grammar checkers, paraphrase apps, and content generators are becoming more widely available, it is critical to investigate their impact on students' writing abilities especially in scientific subjects where clarity, correctness, and structure are critical.

Students. The study might provide students with knowledge on the benefits and challenges associated with AI tools and how to effectively and wisely integrate these technologies into their academic studies that are essential in the field of biology and other scientific discipline, potentially improving their overall academic performance and learning outcomes.

School Administration. The results might provide a guide for administrations in creating policies and support systems that facilitate the responsible and effective use of AI tools, ensuring that students receive a high-quality education aligned with modern technological advancements.

Teacher. The result might benefit the teachers with insights of how AI tools influence student performance and offer awareness for potential over reliance, particularly in technical writing ensuring students harness the benefits of technology that still nurture essential cognitive and analytical skills.

Future Researchers. The result might serve as a foundation for future studies, allowing researchers to explore more specific aspects of AI integration in education. It may provide an opportunity to explore other elements of AI integration, such as its impact on research writing and long-term academic achievement across programs and institutions.

Scope and Limitation of the Study

This study focuses solely on the effects of artificial intelligence (AI) on the technical writing skills of second-year Bachelor of Science in Biology students at Aklan State University. The scope is limited to this specific student population and does not generalize to other programs or universities. The research will utilize a specific set of AI tools and will assess technical writing skills through a defined set of criteria. The timeframe of the study will be limited to the 2nd Semester, Academic Year 2025-2026.

The study's findings are limited by the sample size and the specific AI tools and assessment methods employed. External factors influencing writing skills, such as prior writing experience and individual learning styles, are not directly controlled for in this study. Finally, the study will only examine the immediate impact of AI use and will not track long-term effects on writing skills.

Definition of Terms

The terms employed in this study were defined according to their intended usage in order to promote a tangible comprehension of the research.

Artificial Intelligence (AI). In the context of this study, artificial intelligence refers to the replication of human intellect in computers that have been taught to think and learn similarly to humans, particularly with regard to natural language processing and generation tools used for text production and editing (Smith, 2023).

Technical Writing Skills. According to this study, technical writing abilities are the capacity to accurately, succinctly, and effectively convey difficult scientific and technical knowledge to a particular audience. Organization, clarity, conciseness, accuracy, language, and conformity to standard formatting guidelines are among the critical abilities evaluated (Jones & Davis, 2022).

Effects. The term "effects" in this study refers to the quantifiable influence, modifications, or results seen in the participants' technical writing abilities, such as increases in clarity, decreases in grammatical mistakes, or adjustments in the amount of time needed to do writing assignments.

Aklan State University. The research site for this study is a governmental higher education establishment in Banga, Aklan, Philippines.

Second Year Bachelor of Science in Biology Students. The participants in this study were enrolled in Aklan State University's Bachelor of Science in Biology degree in their second year of study in the 2024–2025 academic year.

Chapter II

REVIEW OF RELATED LITERATURE

This chapter presents the related literature and findings from previous studies. It is divided into three parts: (1) Artificial Intelligence in Education, (2) Artificial Intelligence in Future Career Choice, and (3) Synthesis.

Artificial Intelligence in Education

According to Edtech (2020), the rapid advancement of Artificial Intelligence (AI) is transforming education at all levels, presenting both opportunities and challenges for student learning and development. While Holmes et al. (2019) suggest that artificial intelligence holds the potential to revolutionize education and address diverse learning needs, Castaneda & Selwyn (2018), caution that simply adopting artificial intelligence technologies does not guarantee positive educational outcomes. This highlights the crucial need for a careful and nuanced approach to artificial intelligence implementation and its impact on student learning.

According to Hennekeuser et al. (2024), AI-powered systems can adapt educational content to meet the unique needs of individual students, potentially enhancing learning effectiveness. This approach is particularly relevant to technical writing, where precise feedback is essential for developing strong writing skills. However, a more comprehensive understanding is needed regarding the impact of artificial intelligence tools in technical writing on academic performance within a specific discipline, such as biology. The utilization of artificial intelligence tools affects the students writing proficiency and academic success.

Pursuant to Tobias Schmohl, Alice Watanabe, Nadine Fröhlich and Dominikus Herzberg (2020), gaining academic writing abilities is a crucial component of higher education. It is clear across various fields that there is a significant need among students for introductory and support services related to scientific methods. A nationwide representative study conducted by the German Centre for Higher Education Research and Science Studies (DZWH) indicates that university students encounter challenges when composing papers. 25.7% of this sample even indicate significant challenges (see DZHW, 2018). Over a quarter of all participants (30.4%) consider the current introductory courses on the methods of scientific work to be not very beneficial for their studies.

As the number and diversity of students rise, they often receive less personalized assistance or individual guidance while working on their thesis. Consequently, this exacerbates the issue, resulting in inadequate scientific education and decreasing the students' motivation.

Artificial Intelligence in Future Career Choice

In accordance with Susan J Ely and Milad Rezvani Rad (2024), the application of artificial intelligence (AI) has presented several opportunities as well as challenging problems for engineering students and teachers. For students and teachers looking to use AI to improve learning in the classroom, the possibilities of employing AI-powered advanced tools to create well-structured, cohesive, and captivating essays in higher education institutions are both exciting and potentially challenging. Students can benefit from AI's unique features, which include real-time evaluation, grammar and style recommendations, and help with content creation. AI may also evaluate students' writing styles and give immediate feedback on areas that require more work and adjustment. This can result in an iterative and self-improving process that will help future engineers advance their careers.

As stated by Stina Westman, Janne Kauttonen, Aarne Klemetti, Niilo Korhonen, Milja Manninen, Asko Mononen, Salla Niittymäki and Henry Paananen (2021), Creating accessible services that connect education and job services is a problem for career counseling in the age of lifelong learning. The use of artificial intelligence to assist with advice in both higher education and the workplace has received relatively little attention up to this point.

According to Nosipho Carol Mavuso, Nobert Jere and Darelle vanGreunen (2023), Fast technological advancements have allowed users to receive support and guidance in making decisions. An illustration of this is how university students can utilize technology to investigate various career paths and make knowledgeable choices regarding their future. Despite the growing application of technology overall, the resources for career guidance and tailored career suggestions in South Africa remain constrained. Several constraints exist, including the persistent issue of restricted access to technology, language barriers, and cultural differences that are common in rural regions.

According to Abdulmajeed Bin Dahmash, Mohammed Alabdulkareem, Aljabriyah Alfutais, Ahmed M Kamel, Feras Alkholaivi, Shaker Alshehri, Yousof Al Zahrani, Mohammed Almoaiqel (2020), the concern that AI could replace radiologists in the future negatively affected medical students' interest in pursuing radiology as a profession. Academic radiologists are urged to inform their students about AI and its possible effects while students evaluate radiology as a lifelong career option. The swift progress of AI in radiology will undoubtedly affect the field; however, the apprehension surrounding AI's influence on radiology negatively impacted our participants, making investment in AI education strongly advised.

Synthesis

Several studies suggest that AI in technical writing skills and its impact on academic performance can be viewed as a transformative integration of technology to enhance the writing capabilities of students. By combining the advantages of AI tools in technical writing with the academic needs of Biology students, a novel approach can emerge. The synthesis of AI and technical writing skills can potentially result in improved content generation, refined documentation, and user-friendly materials, which, in turn, may positively influence the academic performance of Biology students.

This synthesis has the potential to streamline the writing process, enhance accuracy, and offer new opportunities for students to engage with complex writing tasks efficiently. Moreover, the integration of AI in technical writing can empower students to create high-quality, error-free content, leading to enhanced academic outcomes and performance for 2nd Year Bachelor of Science in Biology students at Aklan State University. By synthesizing the benefits of AI in technical writing with the academic needs of Biology students, this integration can create a conducive environment for learning, research, and academic success.

The integration of AI in technical writing has been a paradigm-shifting technological advancement, particularly in academic writing (Nature Reviews Urology). AI tools in technical writing offer numerous advantages to writers, such as automated content generation, enhanced accuracy and consistency, real-time documentation assistance, and efficient translation and localization (The Advantages of Artificial Intelligence (AI) in Technical Writing).

Furthermore, AI has revolutionized various industries, including technical writing, by providing precise content generation, suggesting sentence structures, improving readability, and offering user-friendly documentation (The Physiological Society).

The impact of AI on teaching writing skills to life science students has been a subject of intense scrutiny in higher education, with a focus on leveraging AI tools ethically and effectively in learning experiences, The Physiological Society. In academic writing, a thorough review of related literature is essential to understand current knowledge, identify gaps, and provide a solid foundation for research (A Beginner's Guide to Review of Related Literature). The review of related literature helps in defining research problems, framing objectives, and hypotheses, ultimately contributing to the quality and integrity of academic work.

CHAPTER III

METHODOLOGY

This chapter presents the research design, locale of the study, participants of the study, data gathering instrument, data gathering procedures, and data analysis.

Research Design

This study utilizes a quantitative descriptive-correlational research design to investigate the effects of artificial intelligence (AI) tools on the technical writing skills of second-year Bachelor of Science in Biology students at Aklan State University. This design is appropriate because it enables the researchers to gather numerical data and analyze patterns in student behaviors, AI usage, and academic performance without manipulating any variables. The descriptive aspect of the study will capture the extent to which students use AI writing tools such as grammar checkers, paraphrasing software, and AI-powered writing assistants. These tools have become increasingly accessible and influential in shaping how students approach academic and scientific writing.

The study will describe how these technologies are being integrated into students' workflow and what writing features are most affected, such as clarity, conciseness, coherence, and audience appropriateness. The correlational component seeks to determine whether there is a significant relationship between the intensity of AI tool usage and students' performance in technical writing assignments.

It will also examine whether AI support leads to improved scientific literacy or fosters dependency and weakens independent writing skills. By using this research design, the study aims to provide measurable insights into the impact of AI on student writing. It will address key concerns in the academic community, such as the potential for academic dishonesty, overdependence on AI tools, and a decline in critical thinking skills. At the same time, it will explore how responsible use of AI can support and improve scientific communication. The findings may help guide school policies, inform teaching practices, and shape curriculum development to ensure AI is integrated into higher education in a meaningful and ethical way.

Locale of the Study

This study will be conducted at Aklan State University, located in Banga, Aklan, Philippines. Specifically, the research focused on the College of Arts and Sciences, where students enrolled in the Bachelor of Science in Biology program are based. Aklan State University is a government-funded higher education institution known for its commitment to academic excellence and scientific advancement, making it a relevant setting for research involving academic writing and technology integration.

The selection of this university as the study locale was based on its active implementation of modern learning tools and the increasing exposure of students to artificial intelligence applications. The second year BS Biology students, the primary respondents of this study, are expected to have developed foundational skills in technical writing and are currently exposed to AI-assisted tools, making them ideal participants for examining the effects of such technologies on their academic performance. Conducting the research within this environment ensures that the findings are grounded in an authentic academic context and reflect the real-life interactions between AI tools and student learning practices.

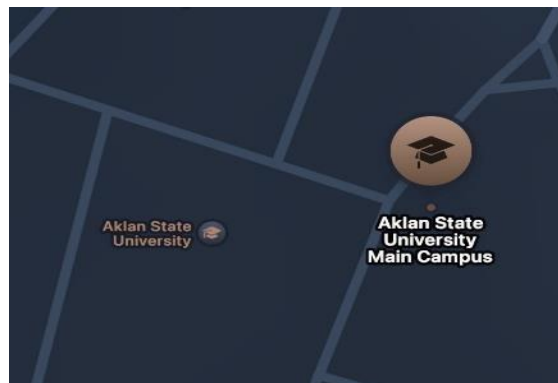


Figure 2. Map showing the location of Aklan State University – Main Campus in Banga, Aklan.

Participants of the Study

The participants of this study will be second-year college students (N = 75) enrolled in the Bachelor of Science in Biology program under the College of Arts and Sciences at Aklan State University, Banga Campus. This group will be selected due to their ongoing engagement with academic writing tasks and increasing exposure to artificial intelligence (AI) tools in their learning environment.

As students navigating the demands of technical writing within a scientific discipline, they are expected to offer relevant and insightful perspectives on the potential influence of AI on writing quality, comprehension, and academic performance. Their active involvement in both academic tasks and emerging technologies makes them suitable respondents for the study.

Moreover, selecting students from the same academic level ensures a more consistent baseline for evaluating how AI tools may support or hinder the development of technical writing skills in higher education.

Ethical Consideration

This study will adhere to the highest ethical standards to ensure the rights, dignity, and well-being of all participants. The purpose, objectives, and procedures of the research will be clearly explained to participants prior to data collection through an informed consent form embedded in the online survey.

Participation in the study will be entirely voluntary. Students will be assured that they may decline or withdraw from participation at any point without any academic consequences. Confidentiality and anonymity will be strictly maintained throughout the research process. No identifying information will be disclosed, and all responses will be securely stored and accessible only to the researchers.

Furthermore, no physical, emotional, or psychological harm will be inflicted on the participants. The data collected will be used solely for academic purposes and will be handled in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173), ensuring responsible data management and protection of privacy.

Data Gathering Instrument

The data to be obtained from the online survey will be analyzed quantitatively, employing both descriptive and inferential statistical techniques. Descriptive statistics, such as frequency, percentage, and mean, will be employed to describe the demographic profile of the respondents and their usage patterns of AI tools in technical writing.

In order to examine the correlation between the frequency of AI tool use and the academic performance of technical writing students, Pearson's correlation coefficient will be used. In this statistical test, it will be established whether a significant correlation is present between AI usage level and academic performance, including writing clarity, coherence, and technical accuracy.

Google Forms survey data will be gathered into a spreadsheet, checked for completeness, and afterward coded for analysis. Statistical analysis will be performed using software like Microsoft Excel. These will allow for precise calculation of results and facilitate the determination of trends and correlations in data. The results will then be interpreted to draw conclusions regarding the effect of AI tools on technical writing as a basis for evidence-based recommendations for enhancing teaching practices as well as student engagement with new technologies.

This study will use an online survey that will employ questionnaires as an instrument to gather data. This type of data gathering will be useful for the researchers to gather honest information and answers from the respondents.

The survey questionnaires will consist of three parts. The first part will be the participant's profile, and the next two parts will be the survey sections.

The first part of the questionnaire will ask for the participant's name, age, sex, and whether they are a second-year BS Biology student at ASU. This section will cover the basic profile information.

The second part of the questionnaire will contain a set of questions and choices for the respondents to answer, presented through checkboxes. It will be designed to identify how often AI tools are used in Technical Writing.

The third part of the questionnaire will also contain a set of questions and choices presented through checkboxes. It will be designed to assess the academic performance of students while using AI tools in writing a technical paper for Technical Writing.

Data Gathering Procedures

This study will survey the second-year college students of the Bachelor of Science in Biology program at Aklan State University – Banga Campus, who meet the qualifications set by the researchers. The researchers will conduct an online survey and will send the questionnaire to the respondents via Facebook Messenger. The respondents will be asked to fill out a Google Form to submit their responses.

Data Analysis

The data to be obtained from the online survey will be analyzed quantitatively using both descriptive and inferential statistical techniques. Descriptive statistics, such as frequency, percentage, and mean, will be used to describe the demographic profile of the respondents and their usage patterns of AI tools in technical writing.

To examine the correlation between the frequency of AI tool use and the academic performance of students in technical writing, Pearson's correlation coefficient will be applied. This statistical test will determine whether a significant correlation exists between the level of AI usage and academic performance, including writing clarity, coherence, and technical accuracy.

Data from the Google Forms survey will be compiled into a spreadsheet, checked for completeness, and then coded for analysis. Statistical analysis will be conducted using software such as Microsoft Excel. This approach will enable precise calculations and will help identify trends and correlations in the data.

The results will then be interpreted to draw conclusions regarding the effect of AI tools on technical writing, serving as a basis for evidence-based recommendations to enhance teaching practices and student engagement with emerging technologies.

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